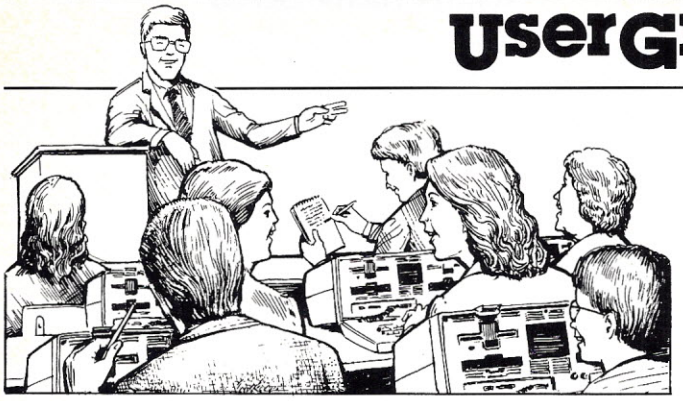
UserGroups



Adding a software printer spooler

John Dilday

Editor's note: this tip is written for the advanced and experienced Osborne user.

My copy of WordStar 2.26 now contains a software printer spooler capability. This feature allows smooth and simultaneous editing and printing of WordStar files. Activating it requires that the "printer busy" connection be active and a subroutine added. Printer busy is part of a standard Centronics type printer connection.

Before activating this feature, do some simultaneous printing and editing by calling ^KP during an edit. Note that editing stops completely when the printer demands control from the processor.

Use INSTALL to complete modifications on a copy of your WordStar:

location	old value	new value	description
0718	00	FF	-HAVSBY busy routine flag
071A	00	C3	JMP -unconditional jump
071B	B7	E0	to unused space in WS
071C	C9	02	user patch areas at 02E0
02E0	00	CD	CALL -call subroutine at
02E1	00	2D	E52D for single density
02E2	00	E5	
02E3	00	A7	ANA -test for busy = FF
02E4	00	C0	RNZ -cond ret if not 00
02E5	00	3F	CMC -comp carry flag to = 0
02E6	00	C9	RET -ret from subroutine

For double density computers, the call address should be at E1Ø3. This address is at 2D lower address byte and the upper address byte is found at location ØØØ2 of your system memory. You may look at ØØØ2 with INSTALL; this value must not be changed. The new byte at location Ø2E2 should be the same as the byte at location ØØØ2 of system memory.

This routine assumes the printer status routine at location 2D leaves the accumulator = Ø if the printer is not busy. In some versions of CP/M this printer status flag is reversed; the byte at location Ø2E4 may have to be changed from CØ to CB.

Locations Ø71A and Ø71C should be the entry address in the WordStar user patch area located at Ø2EØ to Ø35C of your system. Check this area to locate the beginning address of seven consecutive unused bytes. This is the address to use Ø71B lower address byte and Ø71C upper address byte. The bytes at Ø71C and Ø71B will now be Ø2EØ if this is your first patch to the user patch area.

The subroutine was composed by Ted Warn. Inspiration and encouragement came from Erwin Winz. Both are members of R.O.C.C.

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Doubling hard disk memory

Bob Gras

If you had a single-density Osborne computer, but now have a double-density and are using the Corvus hard disk drives, this one is for you.

In the CLINK2K2.ASM file (that is used to produce your CLINK2 program to link you to the Corvus drive), there is a place where the CLINK2 program needs to know a memory offset from CP/M. With the new double density system, more memory is required for managing the Osborne floppy drives, so this value has to be reset to accommodate the double density version of the Osborne. Look for a section in CLINK2K2.ASM that looks like this:

```
     IF ABOVE                ;
                             ;BEGINNING ADDRESS OF CLINK2 *
FREE   EQU       (MSIZE*1024)-CLK2SZ
     ENDIF
     IF NOT(ABOVE)           ;
DELTA  EQU     500H          ; OFFSET FROM STANDARD CP/M *
```

When you find this (it is in the first couple of pages), just type a 900H in place of the 500H. Save the finished new version, then assemble and link the new version. This program will now work with the double density machines.

Not familiar with assembling and linking? So who is? You'll need your CP/M system or utility disk with a copy of ASM.COM and with a copy of LOAD.COM. Then get your Osborne manual and follow the instructions for assembly and linking given on pages 253 through 255 in the version dated "Rev: 2/28/82." If you have a different manual, you may have to dig around a bit.

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